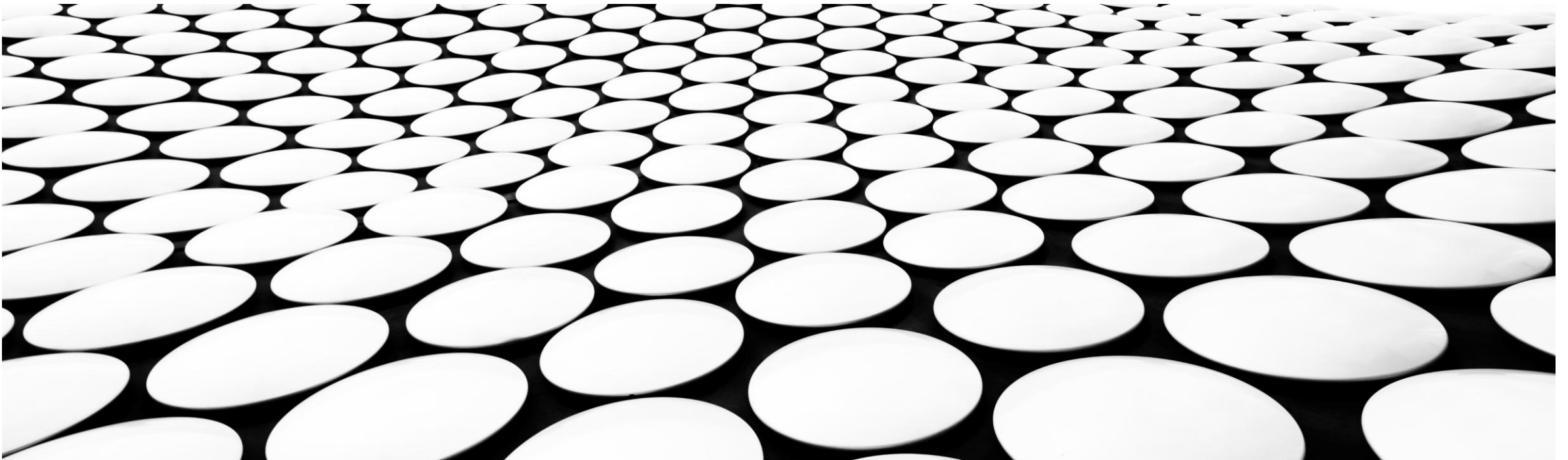

EF2452

MATHEMATICS FOR ECONOMICS AND FINANCE

Lecture 2: Sequences, Limits and Financial Applications

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SEQUENCES

- A sequence is an ordered list of numbers that follows a particular rule. For example, $y_1 = 2, y_2 = 4, y_3 = 6 \dots$
- In economics and finance, a sequence is often used to describe how a variable evolves over time.
- For example, a sequence can be used to describe:
 - the value of an investment over time
 - the size of a population over time

SEQUENCES

- Suppose an employee has a starting monthly salary of \$18,000 and receives a salary increment of \$2,000 each year. Let y_t be the monthly salary in year t .

$$y_0 = 18,000,$$

$$y_t = y_{t-1} + 2,000 \quad (t = 1, 2, 3, \dots)$$

- In this example, the sequence is $\{y_0, y_1, y_2, \dots\} = \{18000, 20000, 22000 \dots\}$

SPECIFYING A SEQUENCE

- A sequence can be specified in different ways:

- Writing out its terms

18,000, 20,000, 22,000

- Using an explicit formula

$$y_t = 18,000 + 2,000t$$

- Using a recurrence equation

$$y_0 = 18,000; y_t = y_{t-1} + 2,000$$

RECURRENCE SEQUENCE

- A **recurrence equation** (or difference equation) defines each term of a sequence using one or more previous terms (y_t in terms of y_{t-1} , y_{t-2} and so on).
- To specify a sequence recursively, we need an initial value and a recurrence equation that determines each subsequent term.

RECURRENCE SEQUENCE

EXAMPLE

There are currently 10,000 doctors in a country. Each year 5% of the existing doctors retire, while 600 new doctors enter the workforce. Let D_t be the number of practicing doctors year t .

The initial value is $D_0 = 10,000$.

The recurrence equation is

$$D_t = 0.95D_{t-1} + 600$$

RECURRENCE SEQUENCE

EXAMPLE

The recurrence equation is

$$D_t = 0.95D_{t-1} + 600$$

This linear recurrence equation uniquely determines D_t from the previous value D_{t-1} .

RECURRENCE SEQUENCE

- A **first-order recurrence** defines the current term y_t using only the previous term y_{t-1} .

- A linear first-order recurrence equation with constant coefficients (a, b) has the form

$$y_t = ay_{t-1} + b$$

- Given an initial value y_0 , the recurrence equation generates a sequence.

RECURRENCE SEQUENCE

- Consider a linear first-order recurrence equation

$$y_t = 2y_{t-1} - 5, \quad t \geq 1,$$

where $a = 2$ and $b = -5$.

- If $y_0 = 0$, then

$$y_1 = 2y_0 - 5 = -5$$

$$y_2 = 2y_1 - 5 = 2(-5) - 5 = -15$$

giving the sequence $\{0, -5, -15 \dots\}$

RECURRENCE SEQUENCE

- Consider a linear first-order recurrence equation

$$y_t = 2y_{t-1} - 5, \quad t \geq 1,$$

where $a = 2$ and $b = -5$.

- If $y_0 = 4$, then

$$y_1 = 2y_0 - 5 = 2(4) - 5 = 3$$

$$y_2 = 2y_1 - 5 = 2(3) - 5 = 1$$

$$y_3 = 2y_2 - 5 = 2(1) - 5 = -3$$

giving the sequence $\{4, 3, 1, -3 \dots\}$

RECURRENCE SEQUENCE

- Different initial values generally generate different sequences.

- If $y_0 = 5$, then

$$y_1 = 2y_0 - 5 = 2(5) - 5 = 5$$

$$y_2 = 2y_1 - 5 = 2(5) - 5 = 5$$

giving the sequence $\{5, 5, 5 \dots\}$.

- The sequence does not change over time. This is called a time-independent solution. $y_t = 5, t = 0, 1, 2, \dots$

FINDING THE TIME-INDEPENDENT SOLUTION

- A time-independent solution y^* of the first-order recurrence equation

$$y_t = f(y_t - 1)$$

is a constant satisfying

$$y^* = f(y^*)$$

- In other words, the constant sequence $\{y^*, y^*, y^* \dots\}$ solves the recurrence equation.

FINDING THE TIME-INDEPENDENT SOLUTION

- Consider the linear first-order recurrence equation

$$y_t = ay_{t-1} + b$$

- For a time-independent solution,

$$y^* = ay^* + b$$

therefore,

$$y^* = \frac{b}{1-a} \text{ (provided } a \neq 1)$$

FINDING THE TIME-INDEPENDENT SOLUTION

EXAMPLE

$$y_t = 2y_{t-1} - 5$$

The time-independent solution:

$$y^* = \frac{b}{1-a} = \frac{-5}{1-2} = 5$$

WHY DO WE NEED A GENERAL SOLUTION

- Consider again the linear first-order recurrence equation

$$y_t = ay_{t-1} + b$$

with an initial value y_0 . We want to find y_t for any t .

- What is y_{100} ? Calculating every term one by one is inefficient.
- We would like to have an explicit formula:

$$y_t = F(t)$$

GENERAL SOLUTION

- The time-independent solution is useful for finding the general solution.

- Consider

$$y_t = ay_{t-1} + b$$

- Since

$$y^* = ay^* + b,$$

subtracting the two equations gives

$$y_t - y^* = a(y_{t-1} - y^*)$$

GENERAL SOLUTION

- Let $z_t = y_t - y^*$. Then

$$y_t - y^* = a(y_{t-1} - y^*)$$

$$z_t = az_{t-1}$$

- Repeated substitution gives

$$z_t = a(az_{t-2}) = a^2 z_{t-2}$$

$$z_t = a(a(az_{t-3})) = a^3 z_{t-3}$$

$$z_t = a^t z_0$$

GENERAL SOLUTION

- Since $z_t = y_t - y^*$ and $z_t = a^t z_0$, we have

$$y_t - y^* = a^t z_0$$

$$z_0 = y_0 - y^*$$

- Therefore,

$$y_t - y^* = a^t (y_0 - y^*)$$

$$y_t = y^* + a^t (y_0 - y^*)$$

- This is the general solution of the linear first-order recurrence equation.

GENERAL SOLUTION

EXAMPLE

Consider again the recurrence equation

$$y_t = 2y_{t-1} - 5 \text{ with } y_0 = 0 .$$

We have found that

$$y^* = 5$$

The general solution:

$$\begin{aligned} y_t &= y^* + a^t(y_0 - y^*) \\ y_t &= 5 + 2^t(0 - 5) = 5 - 5(2^t) \end{aligned}$$

ECONOMIC APPLICATION

National Income Dynamics

- Suppose a closed economy has a national income Y_t in year t , which consists of consumption C_t and investment I_t :

$$Y_t = C_t + I_t$$

- The model assumes that

$$C_t = \frac{1}{2}Y_t$$

$$Y_{t+1} = kI_t$$

where k is a constant

National Income Dynamics

- From $Y_t = C_t + I_t$ and $C_t = \frac{1}{2}Y_t$, we have

$$Y_t = \frac{1}{2}Y_t + I_t$$

and hence

$$I_t = \frac{1}{2}Y_t$$

ECONOMIC APPLICATION

National Income Dynamics

- Since $Y_{t+1} = kI_t$, we can write

$$Y_t = kI_{t-1}$$

- Then,

$$I_t = \frac{1}{2}Y_t = \frac{k}{2}I_{t-1}$$

- We have $a = k/2$ and $b = 0$.
- Therefore, the investment follows a first-order recurrence equation with constant coefficients.

ECONOMIC APPLICATION

National Income Dynamics

- We have

$$I_t = \frac{k}{2} I_{t-1}$$

where $a = k/2$ and $b = 0$.

- The time-independent solution satisfies

$$I^* = \frac{k}{2} I^*$$

- Hence, $I^* = 0$ (provided $k \neq 2$).

National Income Dynamics

- Using the general solution,

$$I_t = I^* + a^t(I_0 - I^*)$$

$$I_t = \left(\frac{k}{2}\right)^t I_0$$

- This gives investment in any year t , given the initial investment I_0 .

ECONOMIC APPLICATION

National Income Dynamics

- Investment rises from year to year if

$$I_{t+1} > I_t$$

$$\left(\frac{k}{2}\right)^{t+1} I_0 > \left(\frac{k}{2}\right)^t I_0$$

$$\left(\frac{k}{2}\right) \left(\frac{k}{2}\right)^t I_0 > \left(\frac{k}{2}\right)^t I_0$$

$$\frac{k}{2} > 1$$

- Assuming $I_t > 0$, investment rises from year to year if $k > 2$, investment rises.

LIMITS

- Consider the general solution

$$y_t = y^* + a^t(y_0 - y^*)$$

We are often interested in what happens to y_t as t increases and the result depends on what happens to a_t .

- The **limit** of a sequence is the value that y_t approaches as t approaches to infinity ($t \rightarrow \infty$) and we write this in symbols as

$$\lim_{t \rightarrow \infty} y^t$$

LIMITS

- Consider the following example:

$$y_t = \left(\frac{1}{2}\right)^t, \quad y_0 = 1, y^* = 0$$

What happens to y_t as $t \rightarrow \infty$

- The above sequence is

$$\left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} \dots\right\}$$

The terms get closer and closer to 0 ($\lim_{t \rightarrow \infty} y^t = 0$)

LIMITS

- Now, consider:

$$y_t = 2^t, y_0 = 1, y^* = 0$$

What happens to y_t as $t \rightarrow \infty$

- The above sequence is

$$\{1, 2, 4, 8, 16, \dots\}$$

The terms increase without bound $y_t \rightarrow +\infty$

LIMITS

- Consider

$$y_t = 2(-1)^t,$$

What happens to y_t as $t \rightarrow \infty$

- The above sequence is

$$\{2, -2, 2, -2, \dots\}$$

The sequence oscillates between 2 and -2 and does not approach a single value. Hence, the limit does not exist.

LIMITS

- Consider

$$y_t = -3y_{t-1}, \quad y_0 = 1$$

What happens to y_t as $t \rightarrow \infty$

- The above sequence is

$$\{1, -3, 9, -27, \dots\}$$

The sequence alternates in sign and increases in magnitude. Hence, the limit does not exist.

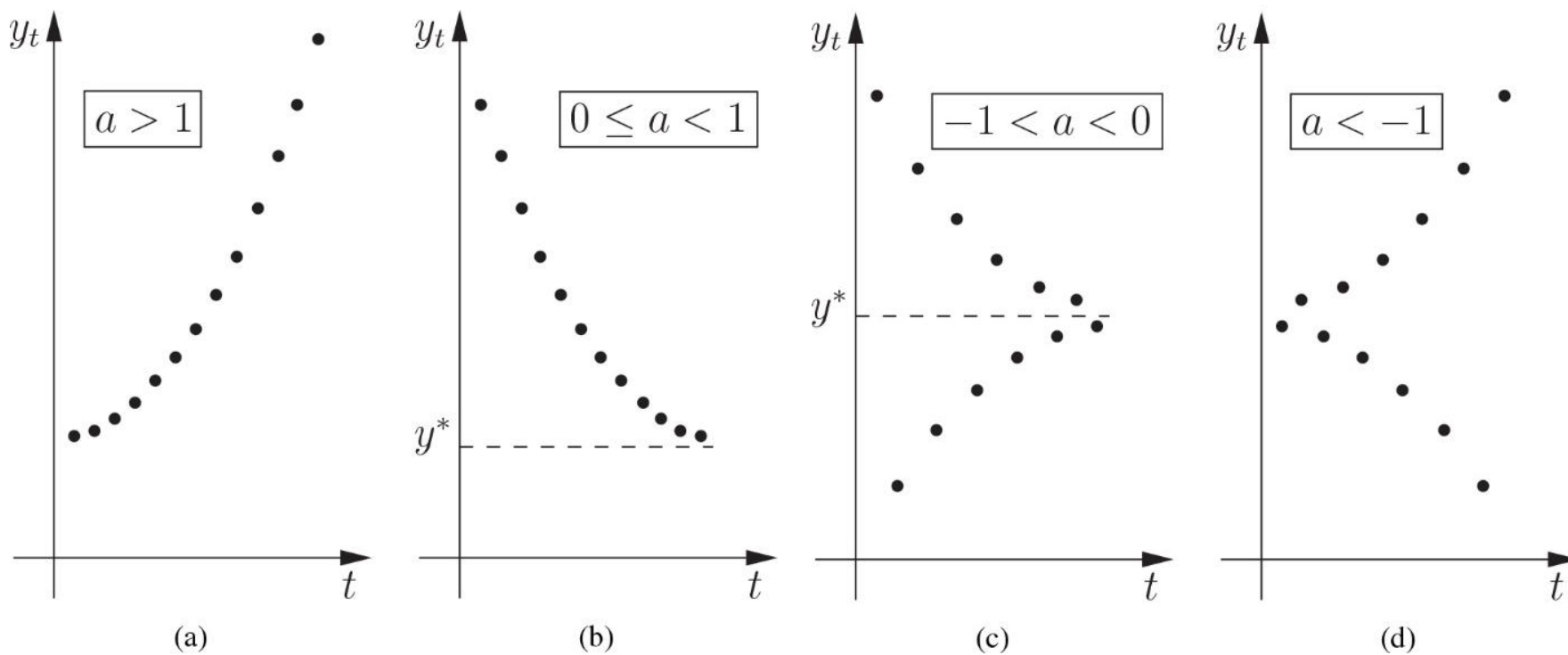
LIMITS

Consider $y_t = y^* + a^t(y_0 - y^*)$ and suppose $y_0 > y^*$.

a	Behavior of a^t as $t \rightarrow +\infty$	Behavior of y_t
$a > 1$	$a^t \rightarrow +\infty$	$y_t \rightarrow +\infty$
$0 \leq a < 1$	$a^t \rightarrow 0$ (decreasing)	$y_t \rightarrow y^*$
$-1 < a < 0$	$a^t \rightarrow 0$ (oscillating)	$y_t \rightarrow y^*$ (oscillating)
$a < -1$	a^t oscillates increasingly	y_t oscillates increasingly

LIMITS

Consider $y_t = y^* + a^t(y_0 - y^*)$ and suppose $y_0 > y^*$.



LIMITS

- If $y_0 < y^*$, the long run conditions are unchanged, but the direction of movement is reversed where appropriate.

LIMITS

- How about the special case when $a = 1$ and when $a = -1$?

- Note that when $a = 1$, the first-order linear recurrence equation is reduced to

$$y_t = y_{t-1} + b$$

- Clearly,

$$y_t = (y_{t-2} + b) + b = y_0 + tb$$

- Therefore, as $t \rightarrow \infty$, $y_t \rightarrow +\infty$ when $b > 0$ or $y_t \rightarrow -\infty$ when $b < 0$ (If $b = 0$, $y_t = y_0$).

LIMITS

- In the case of $a = -1$, the first-order linear recurrence equation is reduced to

$$y_t = -y_{t-1} + b$$

- The above sequence is

$$\{y_0, -y_0 + b, y_0, -y_0 + b \dots\}$$

- Therefore, as $t \rightarrow \infty$, sequence oscillates between two value and generally has no limit.

FINANCIAL APPLICATION

Capital Growth

- Suppose an amount P is invested for N years at a constant annual interest rate r . Let y_t be the value of capital at the end of year t , with $y_0 = P$

- The value of the capital in year t is given by

$$y_t = (1 + r)y_{t-1} \quad t = 1, 2, \dots, N$$

- According to the standard form of a linear first-order recurrence equation,

$$a = (1 + r) \text{ and } b = 0$$

FINANCIAL APPLICATION

Capital Growth

- Using the general solution:

$$y_t = y^* + a^t(y_0 - y^*)$$

$$y_t = \frac{b}{1-a} + a^t \left(y_0 - \frac{b}{1-a} \right)$$

- Substituting $a = (1 + r)$, $b = 0$, and $y_0 = P$ gives

$$y_t = \frac{0}{1 - (1 + r)} + (1 + r)^t \left(P - \frac{0}{1 - (1 + r)} \right)$$

$$y_t = (1 + r)^t P$$

FINANCIAL APPLICATION

Capital Growth

- Therefore, the value of the capital after N years is given by

$$C = (1 + r)^N P$$

- Suppose \$10,000 is invested for 7 years at annual interest rate of 8%. Then

$$C = (1 + 0.08)^7 (10,000) = 17,138.24$$

- The interest is added into the lump sum at the end of each year and this is called compounding.

FINANCIAL APPLICATION

Capital Growth

- Suppose \$10,000 is invested for 7 years at annual interest rate of 8%. Then

$$C = (1 + 0.08)^7(10,000) = 17,138.24$$

- The interest is added into the lump sum at the end of each year and this is called compounding.
- Suppose we want to ensure C after N years, what should be the principal P ?

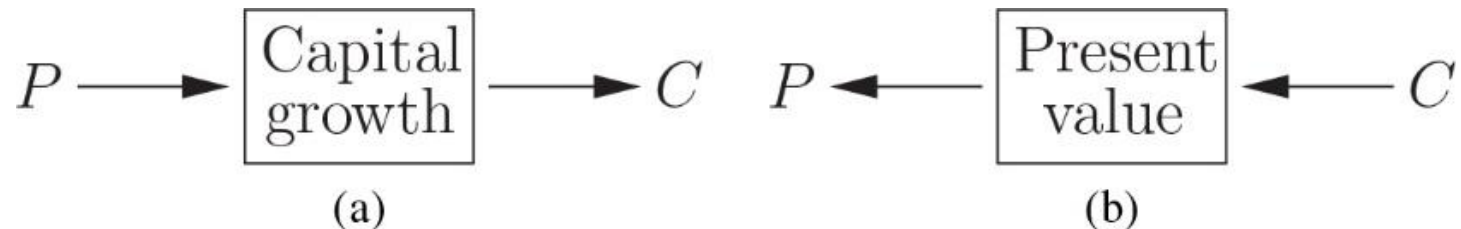
FINANCIAL APPLICATION

Present Value

- Suppose we want to ensure C after N years, what should be the principal P ?
- By simply dividing both sides by $(1 + r)^N$ from $C = (1 + r)^N P$:

$$P = \frac{C}{(1 + r)^N}$$

- P is called the present value of C .



FINANCIAL APPLICATION

Present Value

- Suppose we want to have \$10,000 after 7 years, given the constant annual interest rate of 8%. Then

$$P = \frac{10,000}{(1 + 0.08)^7} \approx 5,834.9$$

- Therefore, \$5,834.9 invested today will grow to \$10,000 after 7 years.

FINANCIAL APPLICATION

Income Generation

- Instead of leaving the capital to grow, suppose an investor invests P and withdraws a fixed amount I at the end of each year. Let y_t be the value of capital at the end of year t , with $y_0 = P$.

- Therefore

$$y_t = (1 + r)y_{t-1} - I \quad t = 1, 2, \dots, N$$

- We have

$$a = (1 + r) \text{ and } b = -I$$

FINANCIAL APPLICATION

Income Generation

- Using the general solution:

$$y_t = y^* + a^t(y_0 - y^*)$$

$$y_t = \frac{b}{1-a} + a^t \left(y_0 - \frac{b}{1-a} \right)$$

$$y_t = \frac{-I}{1-(1+r)} + (1+r)^t \left(P - \frac{-I}{1-(1+r)} \right)$$

$$y_t = \frac{I}{r} + \left(P - \frac{I}{r} \right) (1+r)^t$$

FINANCIAL APPLICATION

Income Generation

- Now assume after N years, the capital is used up.

$$y_N = 0$$

$$\frac{I}{r} + \left(P - \frac{I}{r}\right)(1+r)^N = 0$$

- Rearranging, we have

$$I = P \left(\frac{r(1+r)^N}{(1+r)^N - 1} \right)$$

- This gives the annual income that can be generated from an investment P .

FINANCIAL APPLICATION

Income Generation

- Rearranging for P

$$P = \frac{I}{r} \left(1 - \frac{1}{(1 + r)^N} \right)$$

- This tells us the present value of annuity paying I each year for N years.

FINANCIAL APPLICATION

From an Annuity to a Perpetuity

- Recall that the present value of an annuity is

$$P = \frac{I}{r} \left(1 - \frac{1}{(1+r)^N} \right)$$

- Now suppose the income is received forever. As $N \rightarrow \infty$ and for $r > 0$,

$$\frac{1}{(1+r)^N} \rightarrow 0$$

- Hence, the present value of a perpetuity is $P = \frac{I}{r}$.



ANY QUESTION?